PulseTwin

TEAM GIMITES / DIGITAL TWIN CHALLENGE 2026

Personal health, simulated.

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03 PulseTwin v1.1 · Ten research workbench features

01 / THE PROBLEM

A chart tells only half the story.

01 Daily readings are noisy and disconnected from daily habits.

02 Generic averages hide individual patterns.

03 A useful simulation must expose its assumptions.

02 / SCOPE AND USERS

One focused healthcare use case.

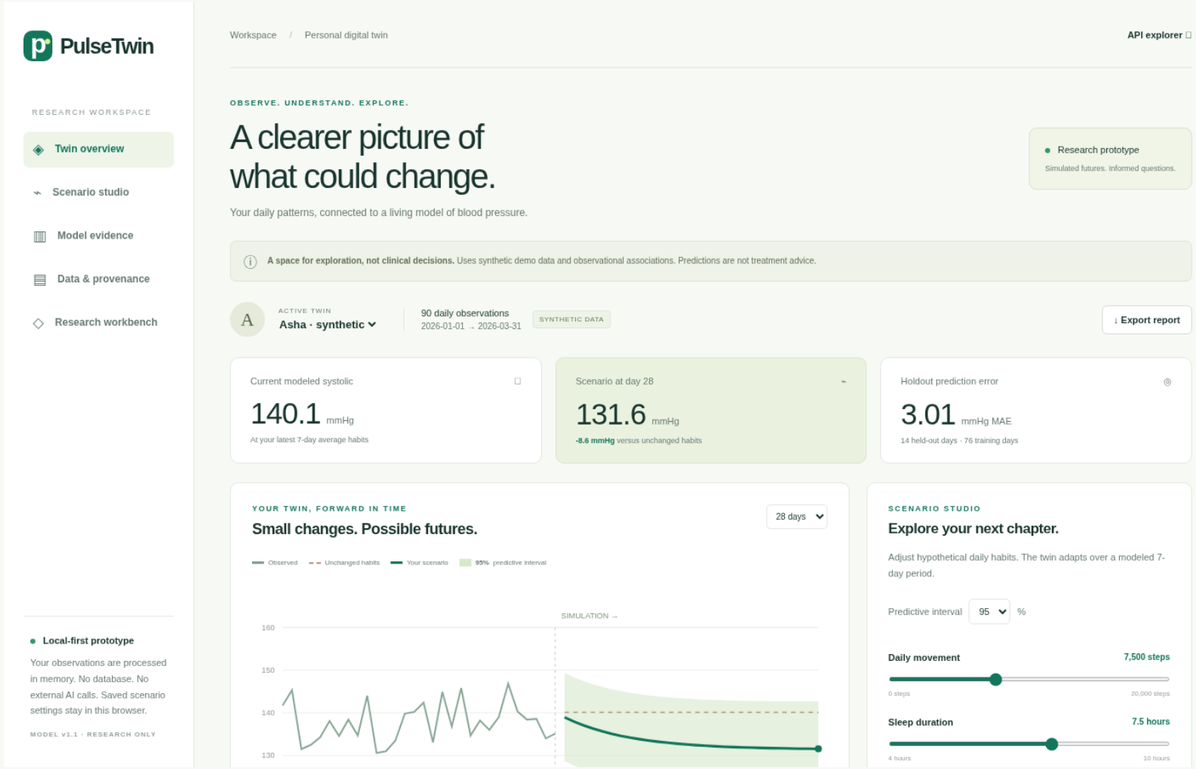
01 Longitudinal cardiometabolic research and education.

02 Systolic blood pressure as the modeled outcome.

03 No diagnosis, medication advice or autonomous decisions.

03 / DASHBOARD WALKTHROUGH

A working research workspace.



01

Three synthetic profiles, each with 90 daily observations.

02

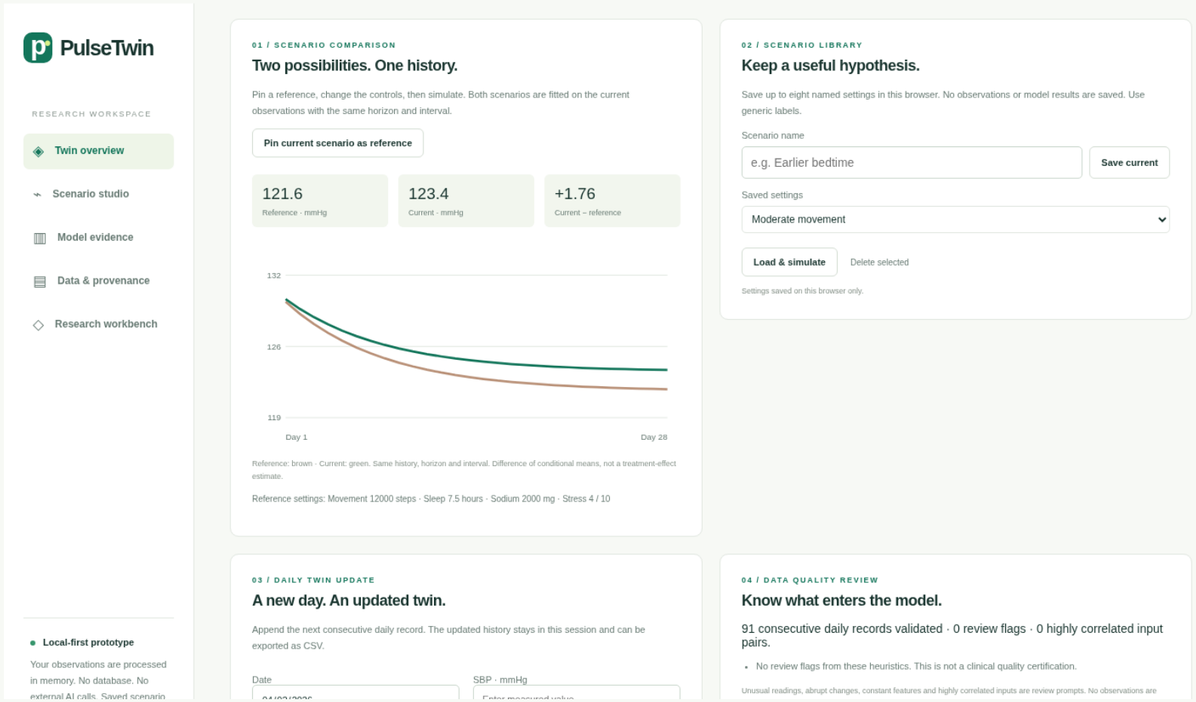
Current modeled SBP, scenario outcome and holdout error.

03

Visible provenance and research-only boundaries.

04 / DAILY UPDATES + DATA QUALITY

A new day. A reviewed update.



01

Append the next daily record and refit the personal twin.

02

Review unusual readings, daily jumps and correlated inputs.

03

No silent deletion, imputation or correction.

05 / BAYESIAN PERSONALIZATION

A model you can inspect.

01 Four standardized habit features plus a personal intercept.

02 Gaussian priors stabilize small histories.

03 Posterior coefficients and covariance use closed-form NumPy algebra.

06 / THE LATENT STATE

Updating the twin over time.

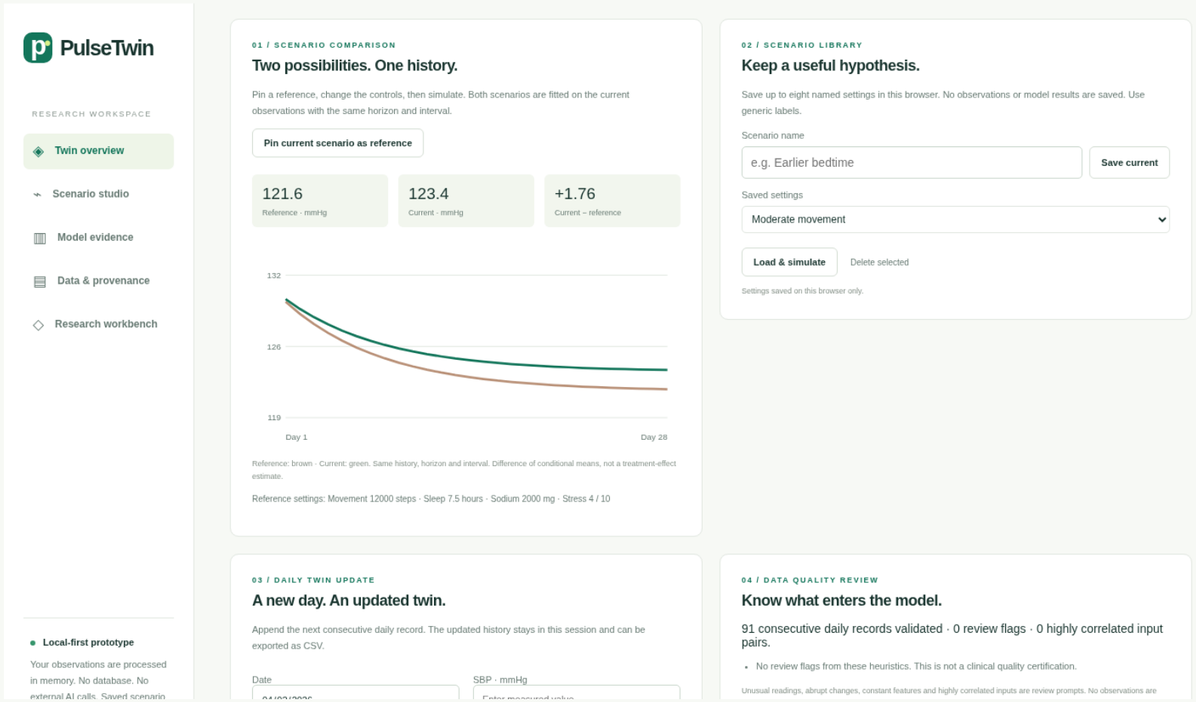
01 Daily residuals update a scalar Kalman state.

02 New observations change the personal fit and current state.

03 Human-in-the-loop prototype; no live device synchronization.

07 / COMPARISON + SCENARIO LIBRARY

Compare and keep useful hypotheses.



01

Pin a reference; compare two conditional trajectories.

02

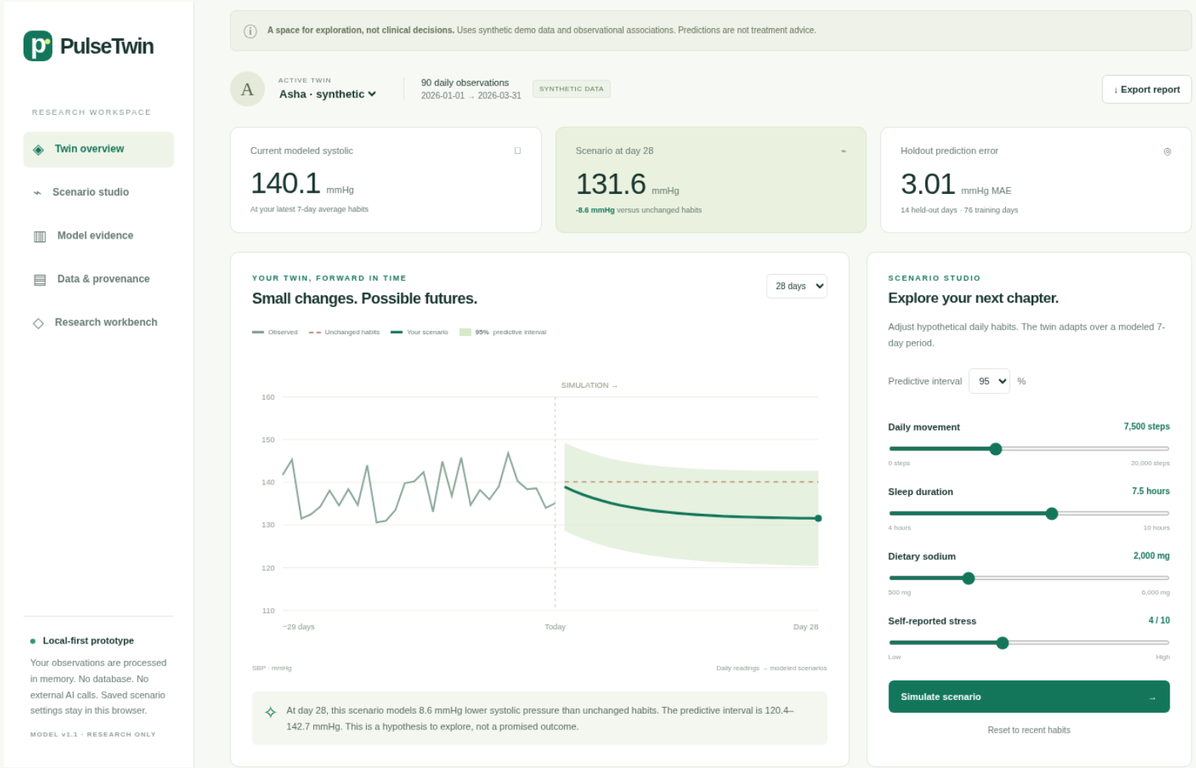
Both fits use the same history, horizon and interval.

03

Save up to eight named settings locally; observations stay out.

08 / ADJUSTABLE PREDICTIVE INTERVALS

Choose the lens. Keep the uncertainty.



01

Choose 80%, 90%, 95% or 99% coverage under the model.

02

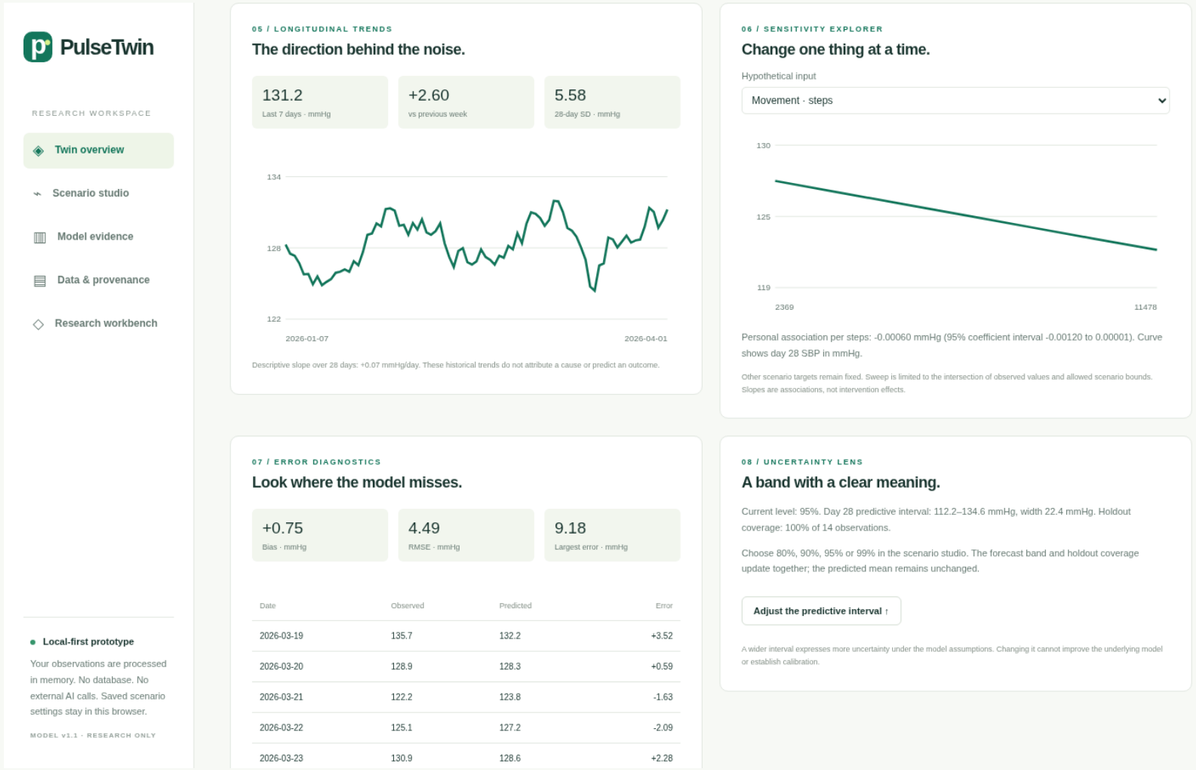
Forecast band and held-out coverage update together.

03

Changing the level never changes the predicted mean.

09 / SENSITIVITY EXPLORER

Change one input. Inspect the response.



01

Seven target values, with other scenario settings held fixed.

02

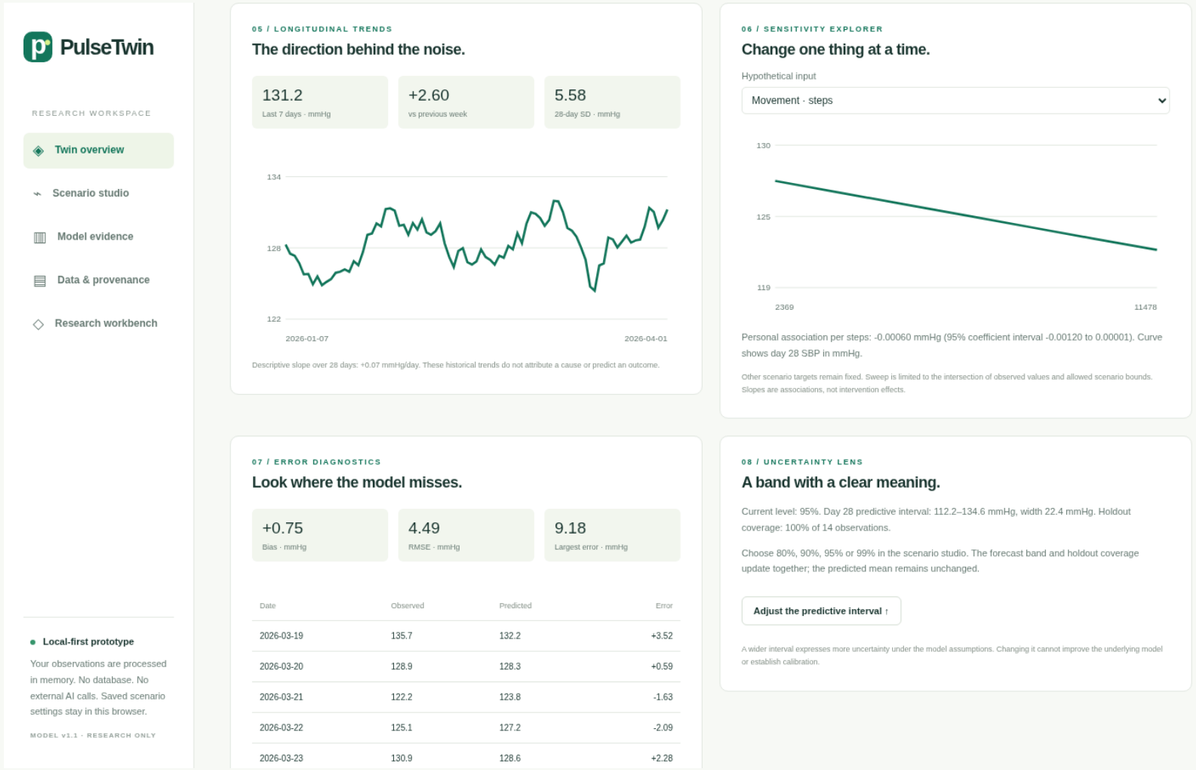
Sweep only within observed and permitted input ranges.

03

Separate 95% coefficient intervals expose association uncertainty.

10 / TRENDS + ERROR DIAGNOSTICS

See the trend. Inspect the misses.



01

Seven-day rolling means, weekly change and recent variability.

02

Holdout residuals, signed bias, RMSE and largest error.

03

Chronological fitting keeps held-out BP labels out.

11 / RESULTS WITH LIMITS

Measured prototype outcomes.

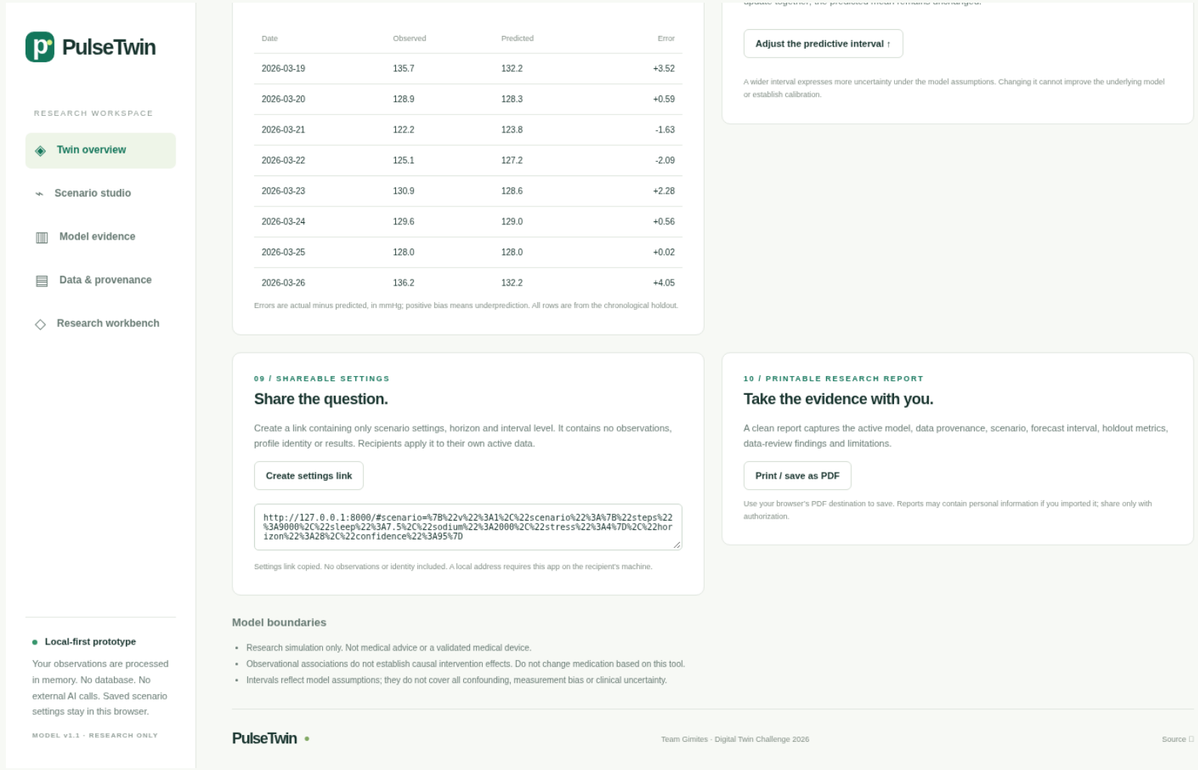
01 Twin MAE: 3.01 / 3.19 / 2.81 mmHg across the three fixtures.

02 Training-mean MAE: 3.60 / 3.85 / 4.47 mmHg.

03 Synthetic data only; 14 test observations per profile.

12 / SETTINGS LINKS + PRINTABLE REPORTS

Share the question. Export the evidence.



01

Share only hypothetical settings in a validated URL fragment.

02

No observation history, profile identity or predictions in the link.

03

Print a report with provenance, outcomes, metrics and limitations.

13 / SYSTEM DESIGN

Simple architecture. Real endpoints.

01 Browser UI FastAPI validation NumPy model.

02 Stateless requests; no database or external AI calls.

03 Reproducible fixtures, evaluation and local artifacts.

14 / ENGINEERING EVIDENCE

Validation beyond a screenshot.

01 24 model and API tests pass.

02 Browser checks exercise all ten new workflows and the original app.

03 Settings privacy, daily updates, PDF output and mobile layout tested.

15 / PRIVACY AND SAFETY

Boundaries are part of the product.

01 Synthetic fixtures; no credentials or patient records included.

02 Observations stay in browser and request memory.

03 No diagnosis, treatment recommendation or clinical certification.

16 / ROADMAP

A credible path beyond the prototype.

01 Consented longitudinal data and external evaluation.

02 Calibration, missingness, subgroup analysis and clinician review.

03 Secure device ingestion before any prospective deployment.

17 / REPRODUCTION AND SUBMISSION

Team Gimites. Reproducible by design.

01 Python 3.12 · FastAPI · NumPy · vanilla JavaScript.

02 Source, model card, PDFs, tests and narrated walkthrough.

03 College affiliation and authorized YouTube publication remain outstanding.